

Predicting Employee Retention – Analysis Report

1. Introduction

This report presents the analysis, modeling approach, results, and insights derived from the Employee Retention dataset. The goal of the study is to understand the key factors influencing employee attrition and to build a predictive model that helps the organization take proactive retention measures.

2. Business Objective

Employee attrition leads to increased hiring costs, productivity loss, and knowledge gaps. The objective of this project is to predict whether an employee is likely to leave the organization and to identify the most influential drivers of attrition using a logistic regression model.

3. Data Understanding

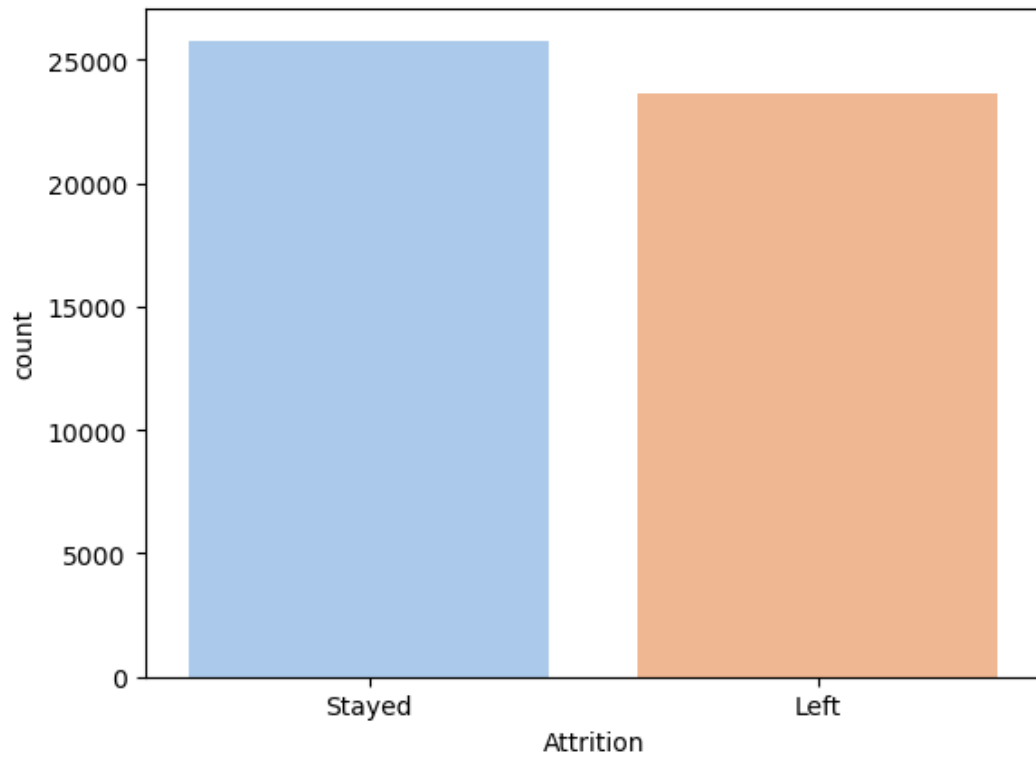
The dataset consists of employee demographic, performance, and workplace-related variables. A detailed data dictionary was used to understand the meaning and relevance of each feature. Both numerical and categorical variables were present in the data.

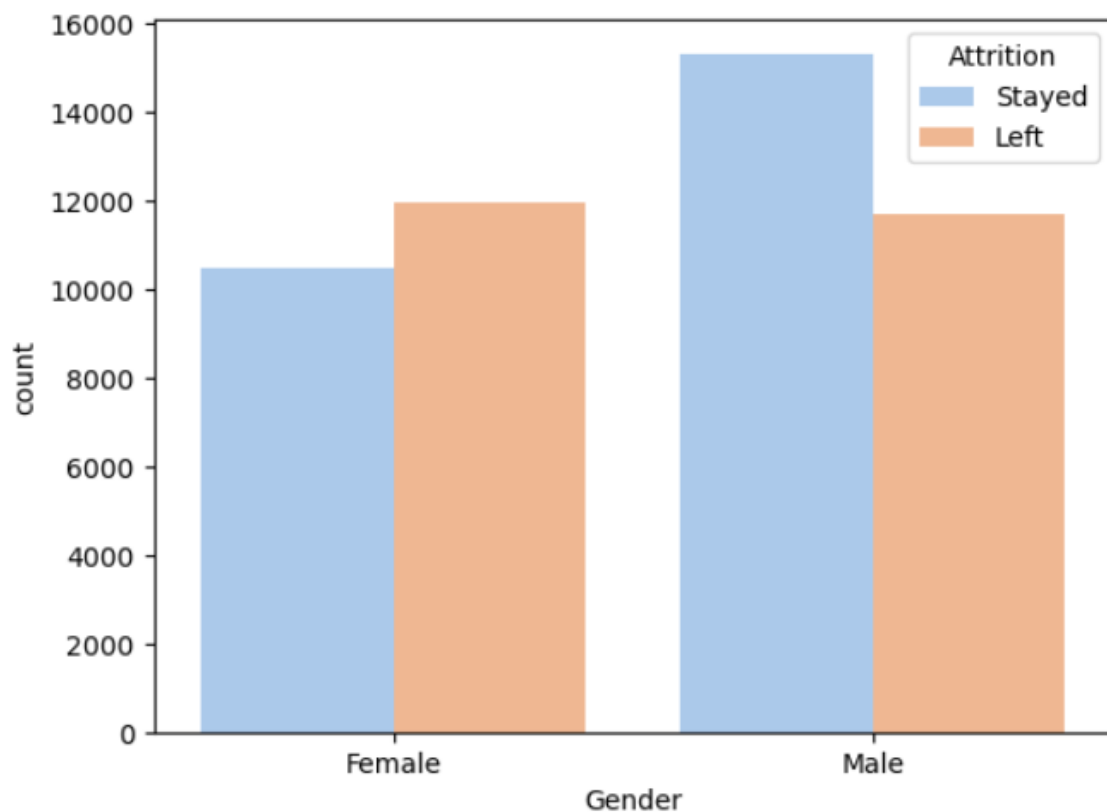
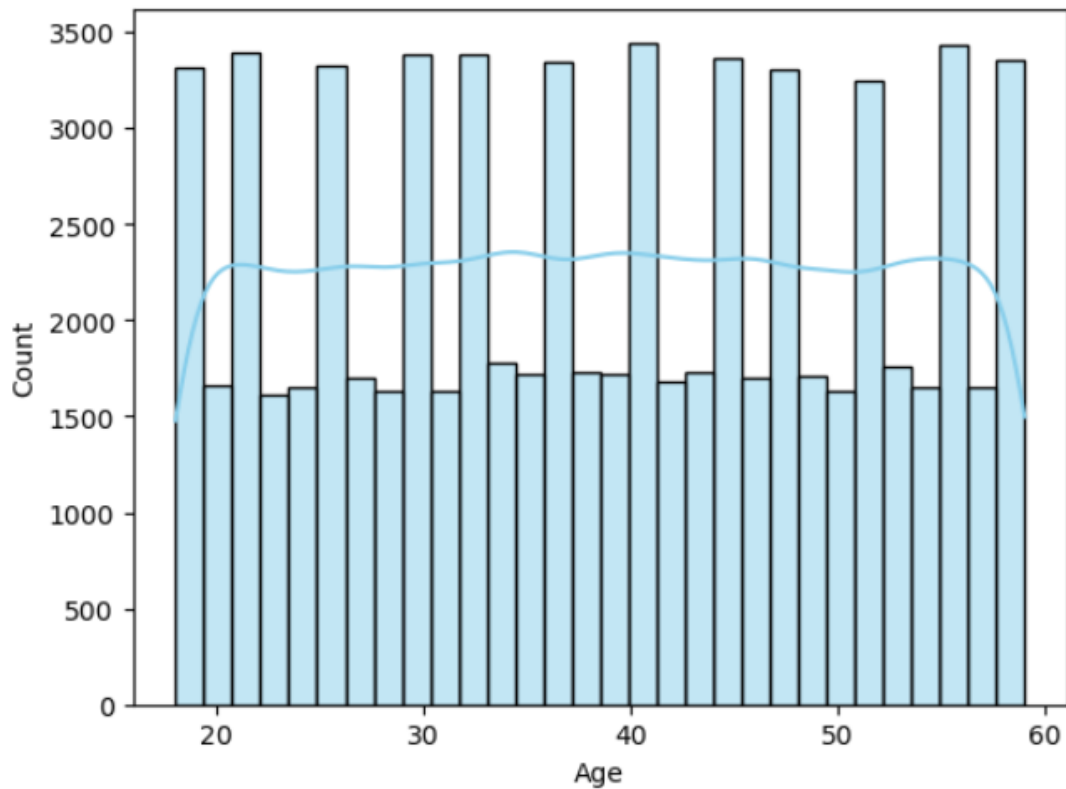
	Employee ID	Age	Gender	Years at Company	Job Role	Monthly Income	Work-Life Balance	Job Satisfaction	Performance Rating	Number of Promotions	...	Number of Dependents	Job Level	Company Size	Company Tenure (In Months)	Remote Work	O
0	8410	31	Male	19	Education	5390	Excellent	Medium	Average	2	...	0	Mid	Medium	89.0	No	
1	64756	59	Female	4	Media	5534	Poor	High	Low	3	...	3	Mid	Medium	21.0	No	
2	30257	24	Female	10	Healthcare	8159	Good	High	Low	0	...	3	Mid	Medium	74.0	No	
3	65791	36	Female	7	Education	3989	Good	High	High	1	...	2	Mid	Small	50.0	Yes	
4	65026	56	Male	41	Education	4821	Fair	Very High	Average	0	...	0	Senior	Medium	68.0	No	

5 rows × 24 columns

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was carried out to identify patterns, trends, and anomalies in the data. Univariate analysis helped understand the distribution of individual variables, while bivariate analysis highlighted relationships between features and employee attrition.





5. Data Preparation

The dataset was split into training and validation sets to ensure unbiased model evaluation. Categorical variables were encoded, and numerical variables were scaled where required. A constant term was added to the feature set before model training.

6. Model Building

A logistic regression model was chosen due to its interpretability and suitability for binary classification problems. The model was trained on the training dataset and used to predict attrition probabilities on the validation dataset.

Logit Regression Results

Dep. Variable:	Attrition_Left	No. Observations:	49444
Model:	Logit	Df Residuals:	49428
Method:	MLE	Df Model:	15
Date:	Tue, 27 Jan 2026	Pseudo R-squ.:	0.2675
Time:	22:54:57	Log-Likelihood:	-25072.
converged:	True	LL-Null:	-34227.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P> z	[0.025	0.975]
const	-0.2356	0.028	-8.336	0.000	-0.291	-0.180
Gender_Male	-0.5738	0.022	-25.875	0.000	-0.617	-0.530
Work-Life Balance_Fair	1.0572	0.025	41.957	0.000	1.008	1.107
Work-Life Balance_Poor	1.2489	0.034	37.262	0.000	1.183	1.315
Job Satisfaction_Low	0.4914	0.037	13.282	0.000	0.419	0.564
Job Satisfaction_Very High	0.4829	0.028	17.462	0.000	0.429	0.537
Performance Rating_Below Average	0.3079	0.031	10.012	0.000	0.248	0.368
Performance Rating_Low	0.5673	0.051	11.058	0.000	0.467	0.668
Overtime_Yes	0.3286	0.023	14.053	0.000	0.283	0.374
Education Level_PhD	-1.4826	0.055	-27.084	0.000	-1.590	-1.375
Marital Status_Single	1.6887	0.025	68.644	0.000	1.640	1.737
Job Level_Mid	-0.9611	0.024	-39.780	0.000	-1.008	-0.914
Job Level_Senior	-2.5326	0.035	-73.125	0.000	-2.600	-2.465
Remote Work_Yes	-1.7154	0.032	-53.379	0.000	-1.778	-1.652
Company Reputation_Fair	0.5296	0.028	18.633	0.000	0.474	0.585
Company Reputation_Poor	0.7359	0.029	25.758	0.000	0.680	0.792

	Features	VIF
0	const	7.182323
1	Gender_Male	1.000454
2	Work-Life Balance_Fair	1.075798
3	Work-Life Balance_Poor	1.075918
4	Job Satisfaction_Low	1.029756
5	Job Satisfaction_Very High	1.029698
6	Performance Rating_Below Average	1.009424
7	Performance Rating_Low	1.009821
8	Overtime_Yes	1.000525
9	Education Level_PhD	1.000210
10	Marital Status_Single	1.000265
11	Job Level_Mid	1.202082
12	Job Level_Senior	1.202138
13	Remote Work_Yes	1.000319
14	Company Reputation_Fair	1.066846
15	Company Reputation_Poor	1.066785

	Attrition_Left	Attrition_Left_Prob	Predicted
0	False	0.622530	1
1	False	0.056655	0
2	True	0.232063	0
3	True	0.563437	1
4	True	0.197320	0

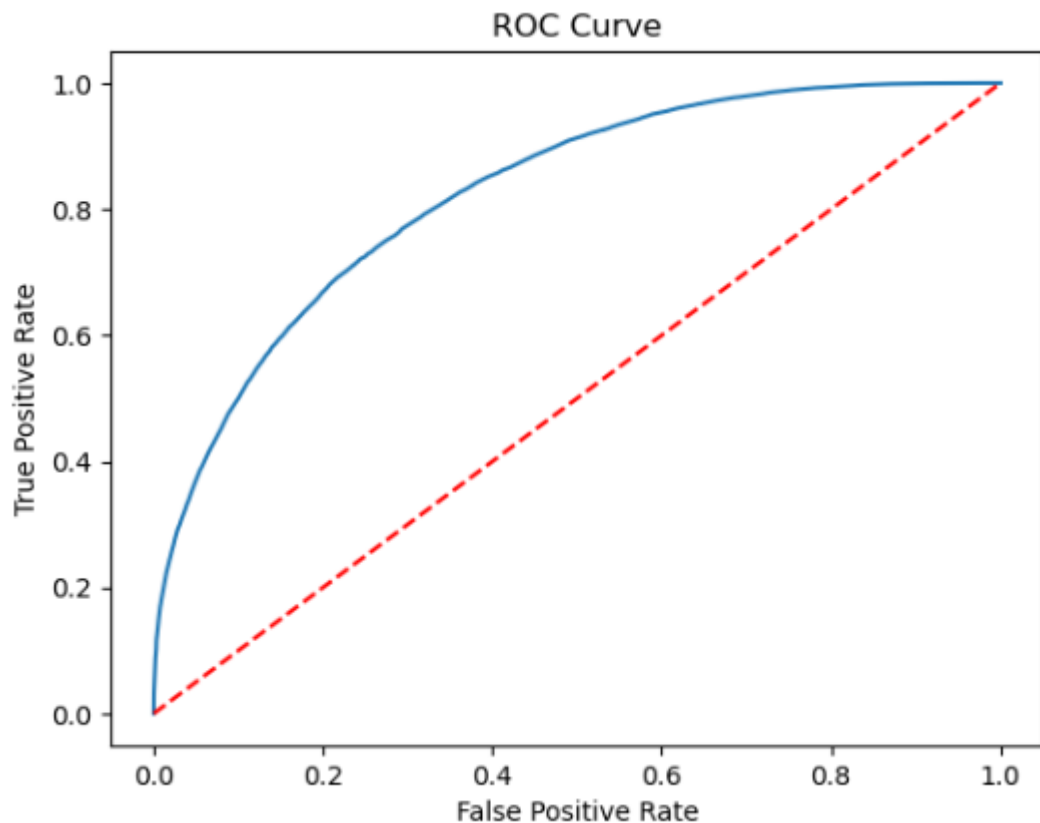
7. Model Evaluation

The model performance was evaluated using multiple metrics including accuracy, confusion matrix, sensitivity, specificity, precision, and recall. These metrics provide a balanced understanding of how well the model predicts employee attrition.

Confusion Matrix:

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array([[5827, 5202],  
       [1020, 9142]])
```

→ The accuracy of the model turned out to be 70.63%



- Sensitivity of the model turned out to be 0.89
- Specificity of the model turned out to be 0.52
- Precision of the model turned out to be 0.63
- Recall of the model turned out to be 0.89

8. Results and Insights

The model demonstrated stable performance on the validation dataset. Certain features showed a strong influence on employee attrition, indicating that both work-related and personal factors play a role in retention. These insights can help HR teams focus on targeted interventions.

9. Business Recommendations

Based on the findings, the organization can design data-driven retention strategies such as improving work-life balance, revisiting compensation structures, and offering clearer career progression paths for employees identified as high-risk.

10. Conclusion

This project successfully demonstrates how predictive analytics can be applied to employee retention problems. With further refinement and additional data, the model can be enhanced and deployed as a decision-support tool for HR management.